

Instructions: All questions are compulsory. Figures on right indicate marks. If any data is missing, is a part of question. Assume suitable data giving reason.

Q.No

- 1 a) Discuss on energy scenario of Indian power system. Comment on constraints in utilization of conventional energy resources. [5]
b) Explain clearly between conventional and non conventional resources. Which one is best? Why? [5]
- 2 a) Stating different losses in indirect heating derive expression for energy efficiency of an electric furnace. [5]
OR
Discuss on design aspects of energy efficient motors. Derive necessary condition for maximum efficiency in DC motor. [5]
b) A piece of 12 cm x 12 cm x 3 cm insulating material is to be heated by dielectric heating at a frequency of 20 MHz. If the power absorbed is 450 W, calculate voltage necessary for heating and current that flows in the material. Assume $\epsilon_r = 3$, $\tan \delta = 0.05$.
Give classification of different air conditioning units used in industry. [5]
c) Write a brief note on infrared heating [2]
- 3 a) Differentiate clearly between preliminary and detailed energy audit. Enlist different steps involved in detailed energy audit. [6]
OR
What do you understand by flow diagram of a process? How it is helpful in energy audit study? Explain with suitable example. [6]
b) Explain the significance of ABC analysis in energy audit. [3]
c) Explain energy conservation measures in an electric motor. [3]
- 4 a) Give classification of lighting schemes. Discuss on factor which influence energy efficiency of lighting system. [6]
b) Derive expression for optimum value of power factor; hence determine the value new power factor when the tariff is Rs 180 per KVA of maximum demand plus a flat rate per kwh. Assume additional cost of condenser as Rs 140/ KVAR. ROI and ROD together taken as 10% [6]

- 5 a) Explain the energy conservation measures in a utility system. [4]

OR

Comment on "one unit of energy saved is equal to two units of energy generated" [4]

- b) Define load balancing. State its advantages. In a typical distribution system a 11 kV feeder has loading of 63A, 72 A and 50 A at RYB phases respectively find saving in losses and % increase in spare capacity of feeder if load is balanced. Assume line resistance is 4Ω . [4]
- c) Explain energy conservation in distribution sector using demand side management. [4]
- 6 a) Explain one energy audit study including hypothetical data, energy calculation, scope for energy conservation, conservation measures and pay back period conducted at domestic sector [6]

OR

Define cogeneration. Enlist their types. With neat diagram explain operation of topping cycle. [6]

- b) A 2.5 MVAR, 11 kV capacitor bank has been commissioned at a 33 kV, 2x5 MVA transformer substation having following data.

	Without capacitor	With capacitor
Peak load	8000 kW	8000 kW
Voltage	10.4 kV	11.4 kV
p.f.	0.846	0.99
Peak current on 11 kV	525 A	409 A
Peak current on 33 kV	175 A	136.3 A

33 kV line length is 35 km and conductor resistance is $0.84 \Omega/\text{km}$, load factor = 0.65.

Find saving in losses and sparing system capacity after installation of capacitor.

Assume $L_s = 0.31l + 0.72l^2$

[4]

- c) With suitable example explain payback period. [2]

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